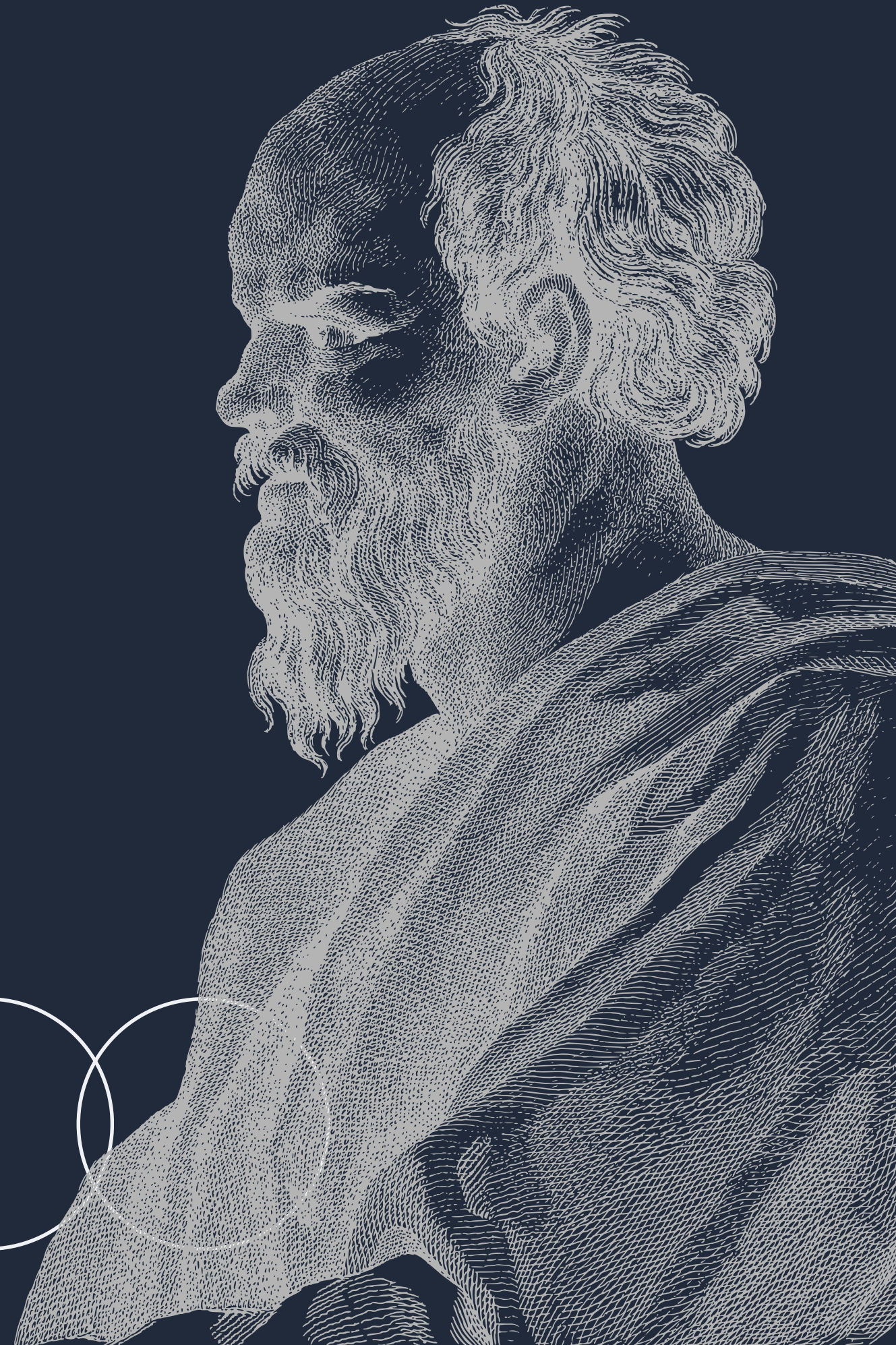


SOCRA PROJECT

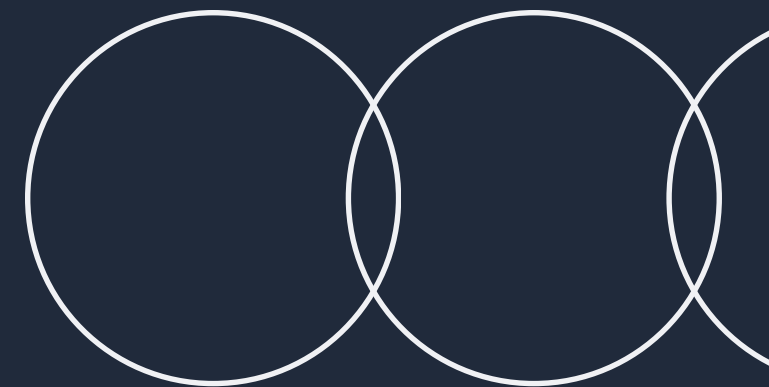
Socra AI- Powered Socratic Tutoring

Presented to Project Mentor



AutoTutor 2005

- Intelligent tutoring via mixed-initiative natural language dialogue
- Guides students to construct answers over 50–200 conversation turns
- Uses LSA (Latent Semantic Analysis) to evaluate responses by meaning, not just word matching
- Follows expectation & misconception tailored dialogue — a computerized marking scheme
- Gives feedback (positive, neutral, negative), asks follow-up questions, drops hints, fills gaps as a last resort
- Relevance to Socra: The foundational proof-of-concept we modernize — RAG and LLMs replace LSA and hand-crafted scripts, achieving the same Socratic structure without manual scripting



ChatGPT Exam Integrity

ChatGPT can pass university exams
with indistinguishable responses

Higher-order thinking

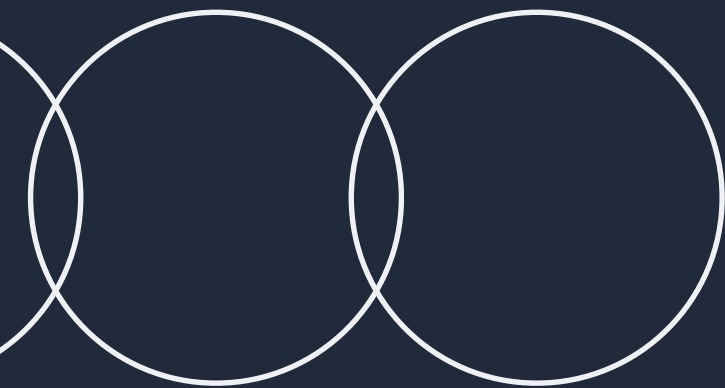
ChatGPT demonstrates the ability to engage in **complex reasoning**, showcasing skills beyond simple memorization and answering based on deep understanding.

9 critical dimensions

The evaluation framework comprises nine critical thinking dimensions, including relevance, clarity, and logic, ensuring comprehensive assessment of student responses.

Relevance to Socra

These critical dimensions will serve as a foundation for Socra's evaluation framework, enhancing the quality of student engagement and feedback.



Retrieval-Augmented Generation

Combining parametric and non-parametric memory methods

RAG Variants

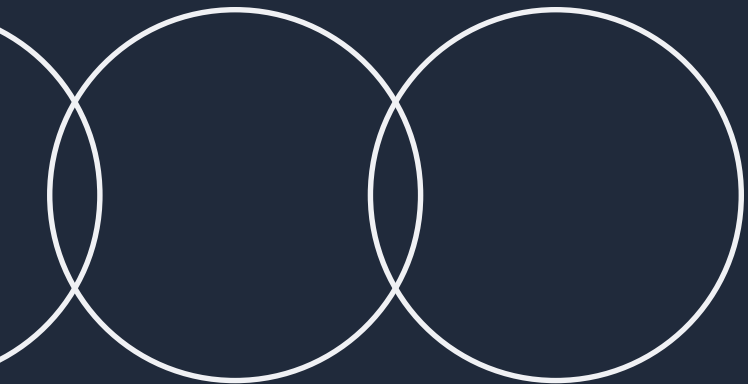
RAG features two variants: RAG-Sequence and RAG-Token, each optimizing different aspects of question answering and document retrieval.

Performance

RAG outperforms BART in open-domain QA, demonstrating significant advantages in both abstractive question answering and question generation tasks.

Hallucination Reduction

Compared to purely parametric models, RAG shows significantly less hallucination, enhancing the reliability and accuracy of generated responses.



Chatting and Cheating

ChatGPT writes academic papers indistinguishable from human work

Plagiarism Statistics

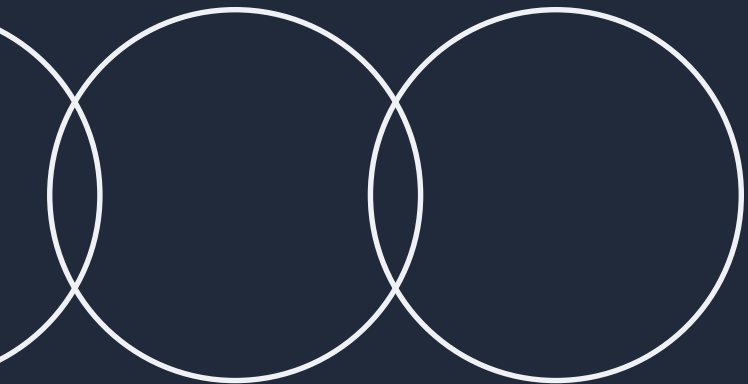
Recent studies show that **14-22 percent** of students admitted to plagiarism even before the advent of ChatGPT and similar technologies.

Detection Tools

While detection tools are available, an **arms race** is emerging between AI-generated content and methods to identify it effectively.

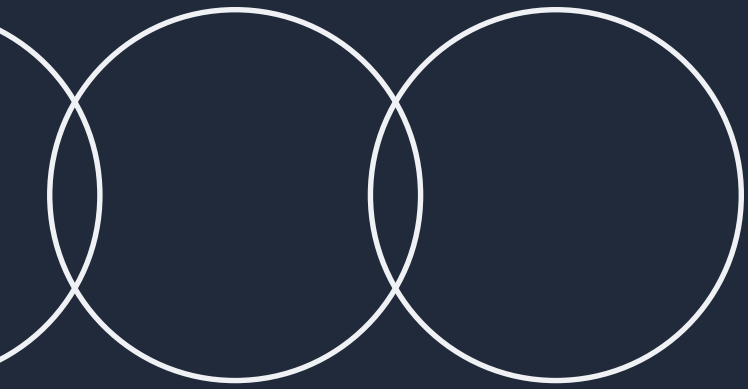
Redesign Assessments

To combat cheating, a better approach is to **redesign assessments** toward critical thinking, dialogue, and presentations to engage students.



Learning Analytics Dashboards

Visualizing learning traces for awareness and change



Impactful Examples

The Course Signals system improved retention rates significantly, demonstrating the power of dashboards in enhancing student engagement and success.

Key Metrics

Essential metrics include time spent on tasks, social interaction levels, and overall assessment performance to inform teaching strategies effectively.

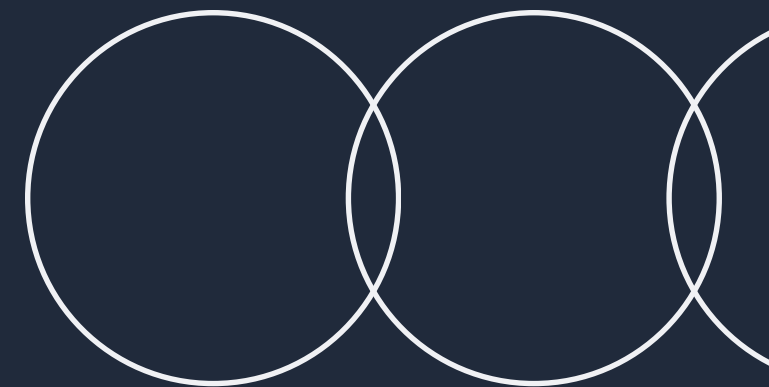
Relevance to Socra

Insights from learning analytics shape our feedback loop design, ensuring we connect multiple data types with goal-oriented progress views.

IRB Requirements

Essential Research Protocol Steps

All research involving Socra requires a Faculty Principal Investigator, completion of CITI Training, a clearly defined Research Protocol, and consent forms from all participants to ensure ethical standards are met.



IRB Detailed Planning

Socra

Timeline

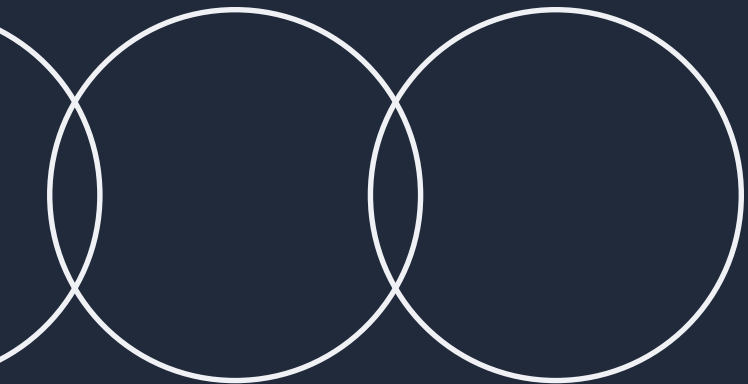
- Week 1: Draft protocol, consent forms, begin CITI Training
- Week 2: Faculty PI review & sign-off
- Week 3: Submit IRB

Data Privacy

- Collects: chat logs, response scores, session duration
- Anonymized, encrypted, accessible only to research team
- No PII retained beyond consent forms

Participants

- Target: 18+ undergraduates in relevant courses
- Voluntary recruitment via course announcements
- Exclusion: minors or students outside target courses



Compute Resources & CCR Request

What Socra Actually Needs:

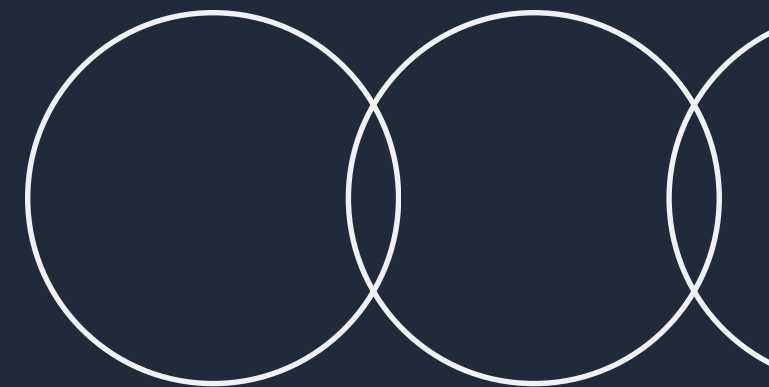
- Vector database for RAG knowledge store (low compute)
- Embedding generation for course materials (moderate compute)
- Fine-tuning a smaller open-source model e.g. Llama 3 (GPU needed)
- Inference & dialogue serving during user testing (moderate GPU)

The Ask:

- CCR cluster access under Faculty PI Prof. Dickson
- Estimated usage: embedding jobs + 1-2 fine-tuning runs
- Not training from scratch, we will have targeted, minimal resource footprint

Why CCR + H100s

- UB's CCR provides academic-grade GPU infrastructure
- H100s enable faster fine-tuning iteration cycles
- Keeps student data on-premise, which aligns with IRB privacy requirements
- No API costs during testing phase, hence more sustainable for a research project



What's Next?

Finalizing Project Proposal

Working on fine-tuning the base model

Finalizing architecture and integrating
evaluation strategies

